Predict the surge before it strikes

1 · Problem

- Task: predict daily river discharge (cubic feet per second) for every station/date in test.csv
- Inputs only: rainfall, max/min temperature, soil moisture + 4 catchment attributes
- No past discharge in test.csv, so river 'memory' must come from weather history
- Score: 40% RMSE + 30% RMSE on each station's top-10% days + 20% MAE + 10% R^2
- Must work on stations never seen in training, and on a CPU-only laptop

Data: the three sources from the brief

- USGS NWIS daily discharge (param 00060): 49 stations, incl. the 5 recommended
- Open-Meteo archive at each gauge: precipitation, Tmax, Tmin, soil moisture 0–7 cm
- Catchment attributes: USGS GAGES-II (StreamStats API was offline) — area, elevation, slope, forest
- Train 2012–2022 (193,438 station-days) | held-out test 2023-01 → 2025-10 (49,361 station-days)
- Cleaning: drop ice/equipment-affected days, USGS no-data codes; never fill missing targets

Model: two CPU LightGBM ensembles

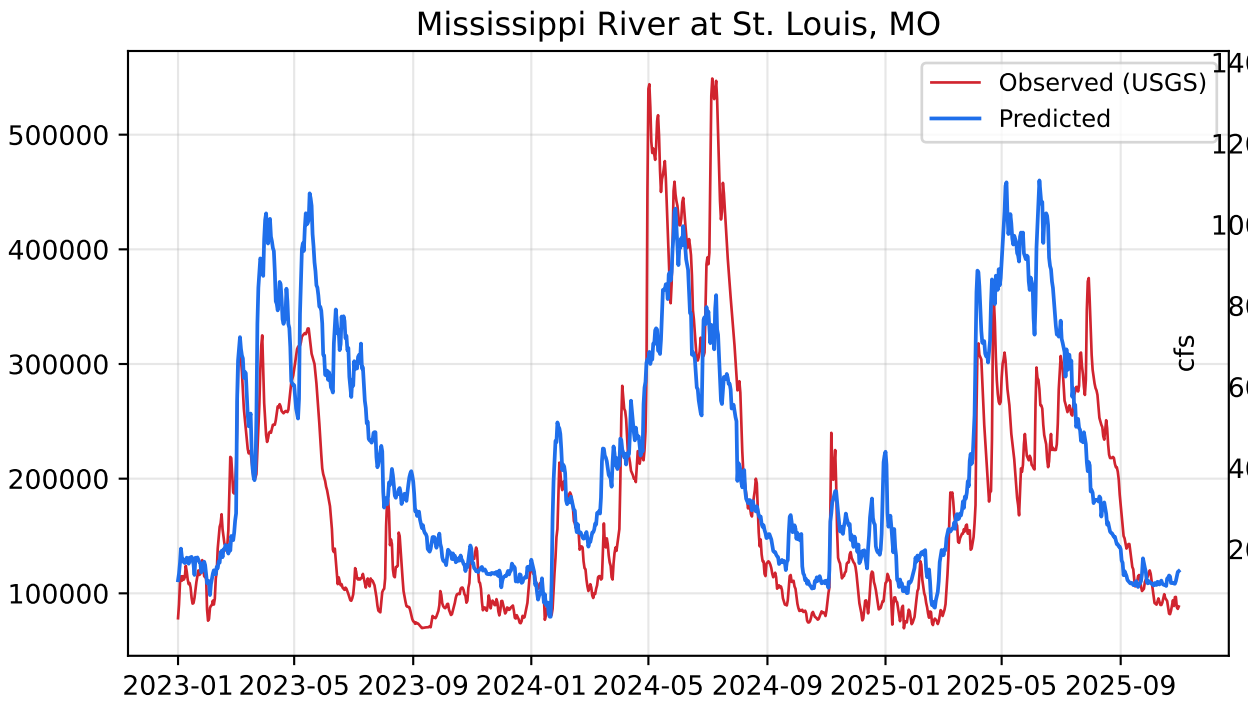
3 · Model

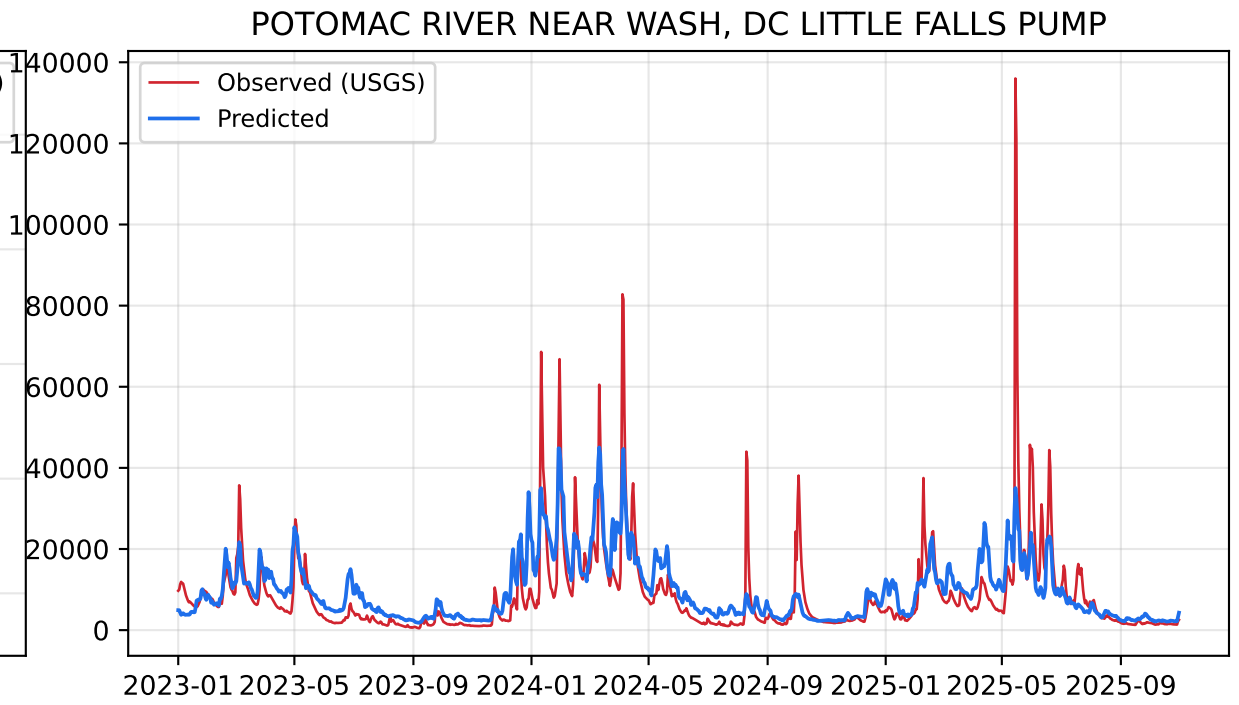
- Features: rolling rain 3–90 d, antecedent-precipitation index, soil trend, snow/melt proxy, season
- Known stations: + station discharge climatology learned from training labels
- Unseen stations: predict discharge per sq mi, multiply by drainage area (extrapolates to big rivers)
- Known-station ensemble averages log-Q, log-Q/area and Tweedie objectives (peaks vs. typical days)
- Every choice made on validation: unseen = log-Q/area alone (R^2 0.74 vs 0.65 for an ensemble)
- Deterministic, 18 MB, trains in ~20 s and predicts 49k rows in ~3 s on a laptop CPU

Results on the 2023–2025 holdout

4 · Results (raw cfs; lower is better except R^2)

	RMSE	RMSE top-10%	MAE	R^2
Monthly climatology baseline	45,913	50,655	12,116	0.815
Our model — known stations	38,807	55,561	8,914	0.867
Our model — unseen stations (CV)	54,290	116,158	15,087	0.741





Limitations & responsible use

5 · Limitations

- Large regulated rivers (Mississippi, Ohio) follow upstream rain and dam releases, not gauge weather
- 2023–2025 were low-water years on the Mississippi; the model over-predicts there
- Dam-controlled rivers (e.g. Colorado at Austin) are poorly captured day-to-day
- Unseen-station accuracy depends on correct drainage area; flashy arid basins are under-represented
- Weather is a single grid point at the gauge, not a basin average
- Research prototype — not an official flood warning; follow local authorities